

unamblify: a real-time neural post-filter for AMBE and Codec 2 digital voice

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<https://rcludwick.github.io/unamblify>

25 September 2026

Abstract

Digital voice on amateur radio inherently operates at a low bitrate. Voice coders transmit a minimal set of parameters that describe speech, from which the receiver reconstructs the voice. As a result, current digital voice transmissions can be challenging to comprehend without prior listener acclimatization.

At 2400 bit/s, the AMBE family [4, 5] discards phase information, inter-harmonic content, frequencies above 3.7 kHz, and temporal structure finer than 20 ms.

unamblify is a neural post-filter designed to operate after the decoder, inferring and restoring these missing components. The model is trained on paired recordings of clean speech and its corresponding degraded output from a vocoder. The dataset includes encodings from a hardware AMBE-3000R chip for D-STAR and YSF/DMR, and the codec2 crate [9] for M17 modes. A Vocos-based [11] synthesizer was fine-tuned specifically for these digital voice modes. UTMOS22 [10] was used as the impartial judge. The resulting model restores a significant portion of the information discarded during encoding. The primary goal of this project is to improve the intelligibility and naturalness of digital voice modes, thereby reducing the cognitive load required to understand transmissions.

1 Introduction

Digital voice modes on amateur radio use a vocoder. A vocoder does not send the sound of your voice. Many times a second it measures a few things about it:

- the pitch,
- how loud each part of the frequency range is,
- whether each part is buzzing (voiced) or hissing (unvoiced).

The receiver reconstructs a new voice from these parameters.

This is how complex speech waveforms can compress into a few thousand bits per second. It is also why digital voice codecs can sound muffled and robotic. The listener hears a reconstructed voice with clarity removed for efficiency.

unamblify runs after the receiver's decoder. It uses two neural networks. The first, the *restorer*, reads the decoded audio and predicts the spectrogram of the original speech. Here a spectrogram is the energy in each of 100 frequency

Table 1: Vocoder bit rates on the air.

Mode	Vocoder	Voice (bit/s)	Total (bit/s)
D-STAR	AMBE	2400	3600
DMR	AMBE+2	2450	3600
YSF (Narrow)	AMBE+2	2450	3600
NXDN (Narrow)	AMBE+2	2450	3600
M17 voice	Codec 2 3200	3200	3200
M17 voice + data	Codec 2 1600	1600	1600

bands, about a hundred times a second. The second network, the *synthesizer*, turns that spectrogram into audio.

The synthesizer is Vocos [11] fine-tuned on ham radio codec audio. Vocos itself was trained on hundreds of hours of recorded speech and the spectrograms of that speech for AI voice synthesis.

The *discriminator* is a third network, used only in training. It tries to distinguish the synthesizer's output from real recordings. The synthesizer was rewarded when it fooled the discriminator.

The key understanding is that Vocos can only render the spectrogram it is given. It cannot add detail that is missing. Given the blurred spectrogram of a codec, it produces a blurred voice. That is why the restorer is used in front of the synthesizer.

1.1 What the codec throws away

Table 1 lists the bit rate of each mode.

AMBE does not try to keep the whole voice. It is based on the multi-band excitation model. Each 20 ms frame carries:

- the pitch (7 bits),
- one voiced or unvoiced decision per band of about 500 Hz,
- the spectral envelope, sampled once per harmonic (57 bits, or 37 in AMBE+2).

Five things are discarded:

1. Phase. The receiver has to make it up.
2. Everything between the harmonics.
3. Everything above 3.7 kHz.
4. Fine voicing detail. Each band gets one voiced or unvoiced bit per 20 ms. Breathly, mixed, and changing sounds are forced to one side.

- Timing inside the frame. Every value is an average over 20 ms, so the start of a consonant is smeared.

Together these cause the familiar digital-voice sound. The author noticed that non-native speakers on AMBE+2 narrow usually took more effort to understand than on a clean recording. This project aims to reduce the cognitive effort by restoring what the codec leaves out.

2 Goals

1. Natural narrowband. Remove the buzz and the robotic sound, and restore the texture of unvoiced sounds and transients, at the vocoder’s 8 kHz rate. The reference for “good” is a clean SSB voice without audible digital distortions.

2. Bandwidth extension. Output at 16 kHz with plausible 4 kHz to 8 kHz content. The reference here is an AM broadcast voice.

3. Real time in Rust. 20 ms frames, streaming, with a low and fixed added latency, and small enough to run alongside the rest of a client’s audio pipeline.

4. Reproducible. Which corpora were used, how they were fetched, how they were pushed through the hardware, and how the model was trained are all scripted and documented.

3 What was learned, in short

This section summarizes the key findings, which are detailed in the remainder of the paper.

For the first attempt a post-filter was implemented. Because the codec discards information, a post-filter can only reshape the remaining signal. The initial model (figure 2) functioned as a filter. The network observed the decoder’s output and adjusted it 100 times per second. It was computationally lightweight, running in a tenth of a millisecond per frame.

Unfortunately, its performance plateaued. Adding more data without increasing the step count degraded performance.

The second model (figure 3) generates a new spectrum before synthesizing the voice. It consists of two interfaced networks. A *restorer* processes the codec output and predicts the original wideband spectrum, generating 100 values every 10 ms with an 85 ms lookahead. The *synthesizer* then converts this spectrum into audio.

Data quality was more impactful than network architecture. Common Voice accounts for 90 percent of the samples, but these volunteer recordings vary in microphone and room dynamics. Training the restorer on the 10 percent of studio-recorded data yielded the most significant improvements.

Table 2 summarizes the current performance. In a blind listening test with 16 held-out speakers across four level-matched systems, the regenerated speech showed notable improvement, often sounding very close to the original recordings. Furthermore, UTMOS22 [10] evaluations indicated a measurable improvement over the raw codec output.

Figure 4 shows the second model on one held-out utterance. The AMBE decoder discards all speech content above

Table 2: 80 evaluation clips were judged by UTMOS22, and 64 clips each from DMR and D-STAR were blindly judged by the author.

	UTMOS22 (1–5)	blind listening test D-STAR	YSF/DMR
original recording	4.14	4.9 [‡]	5.0
raw codec	2.22	1.6	1.9
filter model (first)	2.1 [†]	2.6	2.6
restorer + synthesizer	3.57	4.6	4.6

[†]Measured on an earlier clip set, where the codec scored 1.79 and the recording 4.10.

[‡]Different audio samples were judged with D-STAR and YSF/DMR. Not all original samples were of high quality.

3.7 kHz and flattens the detail between harmonics. The restorer and synthesizer rebuild the upper band and the vocal continuity.

3.1 What needs improvement

Stop consonants are 5 dB to 8 dB quieter than the recording, with the codec’s smear halved; the filter had made them worse than the codec. The “s” is 4.5 dB soft.

4 Generating the training set

Producing the training pairs at scale is most of the effort in the project. While Codec 2 can be parallelized to deliver quick results, the AMBE chip runs closer to real time, so the corpus is bounded by weeks of wall-clock time and not by disk or compute.



Figure 1: The three DVMEGA DVstick 30s that run the AMBE captures. Each holds a DVSI AMBE-3000R chip behind a USB serial link at 460 800 baud, and each runs one capture worker.

4.1 The codecs

Every training pair is clean speech beside the same speech after one real codec. Table 3 lists the codecs used. Each one produces its own capture set, named as the harness names it.

D-STAR through the chip (dstar). The AMBE-3000R is DVSI’s own implementation and the one real D-STAR radios contain, so this set is the reference for the mode. The

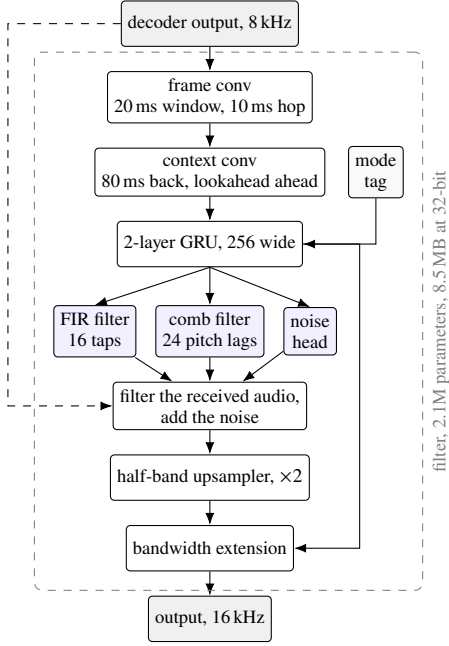


Figure 2: The first model, an adaptive filter in the manner of LACE and NoLACE [3]: 2 124 326 parameters, 8.5 MB of 32-bit weights. Every 10 ms the GRU sets the taps of a short FIR filter, the mix of a pitch comb and, in later runs, the level of a noise source. They are applied to the received audio itself (dashed). A fixed upsampler and a small bandwidth-extension head take the result to 16 kHz.

Table 3: The codecs behind the capture sets. Lag is the decoder’s delay in samples at 8 kHz, measured per set by cross-correlation.

Set	Codec	Implementation	Lag
dstar	AMBE	AMBE-3000R chip	326
ysf-dmr	AMBE+2	AMBE-3000R chip	381
dstar+perens	AMBE	software (Perens)	216
codec2-3200	Codec 2	codec2 crate	140
codec2-1600	Codec 2	codec2 crate	140

chip is set to the D-STAR rate word: 2 400 bit/s of voice in 20 ms frames.

AMBE+2 through the chip (ysf-dmr). The same chip at the AMBE+2 half rate, 2 450 bit/s. System Fusion DN, DMR, and NXDN all use this vocoder mode. YSF DN and DMR carry the same 49 voice bits per frame and differ only in framing and error correction, so one capture serves both.

Bruce Perens’ D-STAR implementation (dstar+perens). The same prepared audio re-encoded by a second, independent implementation of the D-STAR vocoder: the floating-point AMBE path from Bruce Perens’ hams_open [8] (LGPL-3.0-or-later), vendored in the repository. It produces the same bitstream format but not the chip’s audio exactly.¹ This implementation also scores lower on predicted MOS (2.18 against the chip’s 2.56).

Codec 2 at 3200 and 1600 bit/s (codec2-3200,

¹Bruce Perens recently said it was around 97 percent accurate.

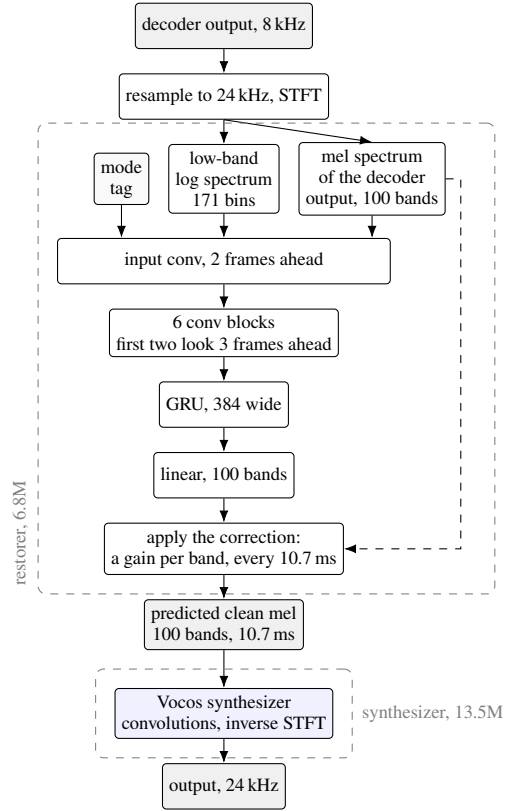


Figure 3: The second model, a restorer and a waveform synthesizer. The restorer predicts a correction to the mel spectrum of the decoder output, before any restoration (the dashed residual path) with 85 ms of lookahead. Vocos, fine-tuned on the restorer’s predictions, turns the corrected spectrum into 24 kHz audio.

codec2-1600). M17’s two voice modes, through the codec2 Rust crate (version 0.3.1, a port of David Rowe’s reference). The 3200 mode uses 20 ms frames and the 1600 mode 40 ms, both 64 bits per frame. Software capture is fast, so these sets cover the whole prepared corpus (section 4).

Lost frames (+drops). A sibling of a set, not a new codec. The stored channel frames of an existing capture have some frames dropped and substituted, then only the decode is re-run. The model sees what a receiver hears when frames go missing on the air.

4.2 The corpus

Table 4 is what has been prepared and how much of it has been through each codec.

4.3 Preparation

Preparation processes each source utterance through a series of steps:

- 1. Decode and resample:** Decode each source file (WAV, FLAC, or MP3) and resample to 16 kHz with a band-limited sinc resampler. A 256-tap Blackman-Harris² windowed sinc (cutoff at 0.95 Nyquist) with group-delay compensation ensures sample-accurate phase and timing alignment.
- 2. Trim silence:** Strip leading and trailing silence below −45 dBFS while keeping a 200 ms margin.

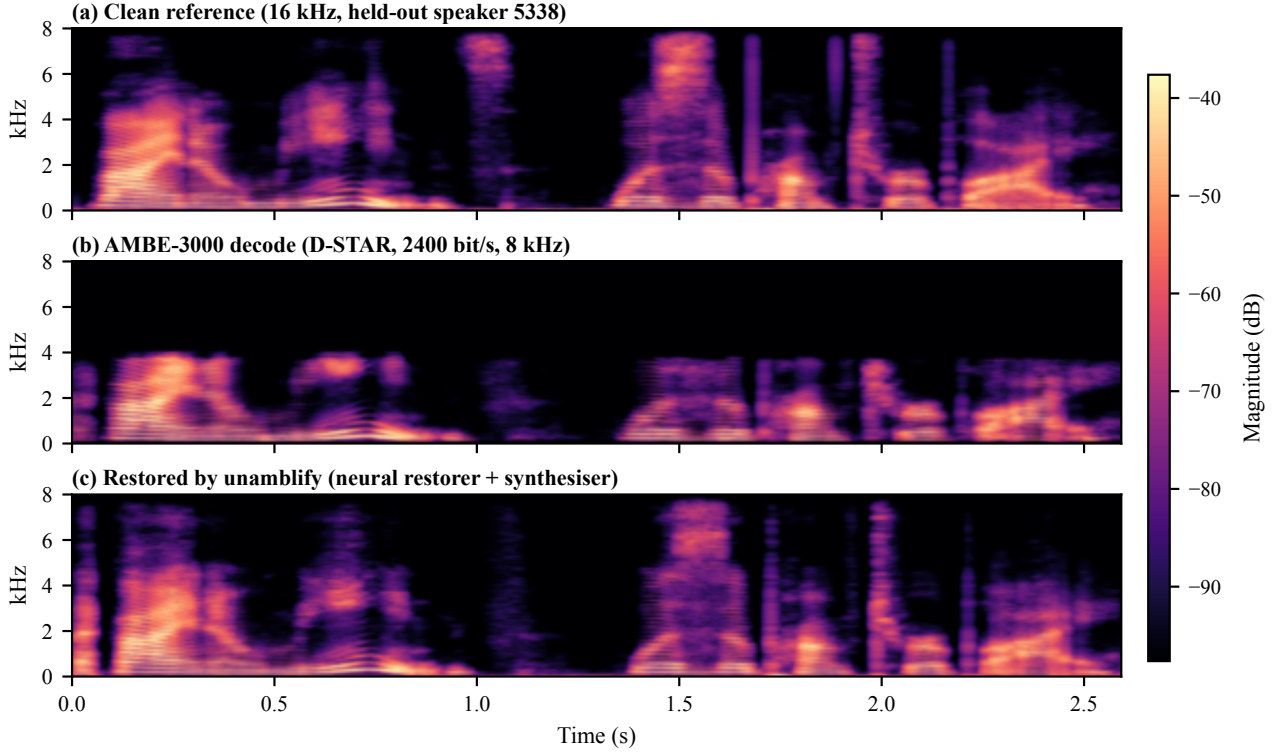


Figure 4: Log-magnitude spectrograms for a held-out evaluation utterance (LibriTTS-R, speaker 5338). (a) Clean 16 kHz studio reference. (b) AMBE-3000R hardware decode (D-STAR mode, 2 400 bit/s, 8 kHz), illustrating the hard cutoff at 3.7 kHz. (c) Restored audio from *unamblify* (neural restorer plus synthesizer), showing reconstructed wideband harmonics up to 8 kHz.

Table 4: Prepared and captured utterances, 25 September 2026.

Corpus	Prepared			Captured			
	Utterances	Hours	Speakers	D-STAR	D-STAR (Perens)	YSF/DMR	Codec 2
Common Voice (en) [1]	1 937 661	2294.4	83 816	309 220	53 987	148 794	1 937 661
LibriTTS-R [7]	347 709	566.5	2 385	152 799	4 182	152 799	347 709
Hi-Fi TTS [2]	257 288	241.0	8	0	0	0	257 288
VCTK 0.92 [13]	44 304	32.0	110	44 304	1 210	44 304	44 304
LJSpeech 1.1 [6]	13 100	23.9	1	13 100	381	13 100	13 100
VoiceBank-DEMAND [12]	12 387	9.4	30	12 387	330	12 387	12 387
Twins	42 571			42 571	42 571	42 571	42 571
Total	2 655 020	3167.2	86 350	574 381	102 661	413 955	2 655 020

3. **Normalize:** Scale active-speech RMS to -26 dBFS.
4. **Filter duration:** Discard any utterance under one second.
5. **Export aligned pairs:** Write two aligned files per utterance: a 16 kHz target signal and an 8 kHz decimated copy for the vocoder.

The corpus is split into “train”, “dev” and “test” groups by speaker, guaranteeing no speaker appears in two splits.

4.4 Augmentation

We add noise to the clean speech and push it through a synthetic overdriven transmit chain to create “twin” utterances. These twins are then used as the input to the codec. Bit losses before the codec and audio distortions before and af-

ter the codec are added. This makes the model more effective for real-world conditions.

4.5 Shards

The final stage joins the capture manifests with the prepared manifest and packs fixed-length training examples into shards, which are flat files the trainer can memory-map.

The filter model’s largest set held 100 000 utterances: 40 000 YSF/DMR, 30 000 D-STAR, and 15 000 each of Codec 2 3200 and 1600. YSF/DMR got the largest share because it is the mode the codec damages most. The restorer is trained on a set drawn only from the studio corpora (LibriTTS-R, VCTK, LJSpeech, VoiceBank-DEMAND and Hi-Fi TTS), with up to 500 000 utterances from each capture set.

5 The two models

Two architectures have been tested. The first is a filter. It reshapes the decoder's output and cannot add what the codec removed. The second predicts the clean spectrum and synthesizes a new waveform from it. Figures 2 and 3 show both, and both are judged by predicted MOS.

Predicted MOS from UTMOS22 [10] is the objective judge. UTMOS22 is trained on human ratings and is not part of the training, so it cannot be optimized against or gamed.

The filter architecture (figure 2) runs in a tenth of a millisecond per frame on one CPU core. Its best predicted MOS was 2.12, against 1.78 for the codec, at step 76 000 of an 80 000-step run, and that turned out to be its ceiling. A probe showed why. Each clip was rebuilt from the filter's output below 3.8 kHz and the clean recording above it, and scored. It scored 2.72, against 2.76 for the codec's own low band in the same mix. A filter can only reshape what the codec kept.

The restorer and synthesizer (figure 3) score 3.57 on MOS, against 2.22 for the codec and 4.14 for the original recordings, measured on the list of captured decodes of the held-out clips.

6 Status and what comes next

The data harness was built to monitor preparation, capture in hardware, and training through a web page. The AMBE captures take the longest time and will run for several weeks, whereas the Codec 2 modes have been completed. The harness is Rust and AGPL-3.0-only, and the filter models also use Rust.

The *restorer + synthesizer* model will need to be optimized for less powerful CPUs. One target is LinHT, the M17 Project's open handheld radio, built on an NXP i.MX 93 processor with two Cortex-A55 cores. It is not yet clear whether that is enough to run the current model in real time.

Further work will be needed to distill the model for embedded CPUs with even further restricted computation and memory requirements.

Biography

Rob Ludwick, AJ7HR. *[Rob has been a licensed amateur radio operator since November 2025, when he received his Extra Class license. He graduated from Iowa State University with a bachelor's degree in computer engineering. He currently works as a software engineer for what vaguely appears to be a large financial institution of some kind or another.]* Callsign page: <https://www.qrz.com/db/AJ7HR>.

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